We consider a one-dimensional scattering problem with an infinite potential wall at $x \leq 0$. A particle of energy $E > 0$ is incident from $x \to +\infty$. In the free region, define

$$q \equiv \frac{\sqrt{2mE}}{\hbar}, \qquad \phi(x) = e^{-iqx} - Se^{iqx}.$$

The incident wave travels towards decreasing x, whereas the reflected wave travels towards increasing x.

I

(1)

For $x > 0$, the wavefunction is

$$\phi_0(x) = e^{-iqx} - S_0 e^{iqx}.$$

Since the potential is infinite for $x \leq 0$, the wavefunction must vanish at the wall:

$$\phi_0(0) = 1 - S_0 = 0.$$

Therefore,

$$\boxed{S_0 = 1}.$$

(2)

Substituting $S_0 = 1$ into the wavefunction,

$$\begin{aligned} \phi_0(x) &= e^{-iqx} - e^{iqx} \\ &= -2i \sin(qx). \end{aligned}$$

Comparing this with $\phi_0(x) = \alpha_0 \sin(qx + \delta_0)$, we identify

$$\boxed{\alpha_0 = -2i, \qquad \delta_0 = 0}.$$

II

(3)

Now the infinite wall extends to $x \leq a$. For $x > a$,

$$\phi_1(x) = e^{-iqx} - S_1 e^{iqx}.$$

The boundary condition is $\phi_1(a) = 0$, which gives

$$e^{-iqa} - S_1 e^{iqa} = 0.$$

Hence,

$$\boxed{S_1 = e^{-2iqa}}.$$

(4)

Using the result of (3),

$$\begin{aligned}\phi_1(x) &= e^{-iqx} - e^{-2iqa}e^{iqx} \\ &= e^{-iqa}\Big(e^{-iq(x-a)} - e^{iq(x-a)}\Big) \\ &= -2ie^{-iqa}\sin\big(q(x-a)\big).\end{aligned}$$

Since

$$\sin\big(q(x-a)\big) = \sin(qx - qa),$$

and $0 < qa < \pi/2$, comparison with $\phi_1(x) = \alpha_1 \sin(qx + \delta_1)$ yields

$$\boxed{\alpha_1 = -2ie^{-iqa}, \qquad \delta_1 = -qa}.$$

Physical interpretation. The negative phase shift indicates that extending the hard wall from $x = 0$ to $x = a$ shifts the nodes of the standing wave towards larger x by a distance a.

(5)

For the reflected-wave region $x > a$, the difference between the two wavefunctions is

$$\begin{aligned}\phi_0(x) - \phi_1(x) &= (e^{-iqx} - e^{iqx}) - (e^{-iqx} - e^{-2iqa}e^{iqx}) \\ &= e^{iqx}\big(e^{-2iqa} - 1\big).\end{aligned}$$

Therefore,

$$\begin{aligned}f_1 &= |\phi_0(x) - \phi_1(x)|^2 \\ &= |e^{-2iqa} - 1|^2 \\ &= (e^{-2iqa} - 1)(e^{2iqa} - 1) \\ &= 2 - 2\cos(2qa).\end{aligned}$$

Using $1 - \cos(2\theta) = 2\sin^2\theta$, we finally obtain

$$\boxed{f_1 = 4\sin^2(qa)}.$$

III

We now replace the extra hard-wall region $0 < x \le a$ by an attractive square well of depth $V_0 > 0$:

$$V(x) = \begin{cases} +\infty, & x \le 0, \\ -V_0, & 0 < x \le a, \\ 0, & x > a. \end{cases}$$

Write

$$Q \equiv \frac{\sqrt{2m(E + V_0)}}{\hbar}, \qquad q \equiv \frac{\sqrt{2mE}}{\hbar}.$$

(6)

Inside the well, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{d^2\phi_{\text{in}}}{dx^2} - V_0\phi_{\text{in}} = E\phi_{\text{in}},$$

or, equivalently,

$$\frac{d^2\phi_{\text{in}}}{dx^2} = -Q^2\phi_{\text{in}}.$$

Its general solution with two arbitrary constants is

$$\boxed{\phi_{\text{in}}(x) = A\sin(Qx) + B\cos(Qx)}.$$

The infinite wall imposes $\phi_{\text{in}}(0) = 0$, so $B = 0$. Consequently,

$$\boxed{\phi_{\text{in}}(x) = A\sin(Qx)}.$$

(7)

In the free region $x > a$, we write

$$\phi_2(x) = e^{-iqx} - Se^{iqx}.$$

Since the potential has only a finite jump at $x = a$, both the wavefunction and its first derivative must be continuous there:

$$\begin{aligned} A\sin(Qa) &= e^{-iqa} - Se^{iqa}, \\ QA\cos(Qa) &= -iq\big(e^{-iqa} + Se^{iqa}\big). \end{aligned}$$

Eliminating A gives

$$Q\cot(Qa)\big(e^{-iqa} - Se^{iqa}\big) = -iq\big(e^{-iqa} + Se^{iqa}\big).$$

Rearranging,

$$\begin{aligned} \big[Q\cot(Qa) + iq\big]e^{-iqa} &= \big[Q\cot(Qa) - iq\big]Se^{iqa}, \\ S &= e^{-2iqa}\frac{Q\cot(Qa) + iq}{Q\cot(Qa) - iq} \\ &= e^{-2iqa}\frac{iQ - q\tan(Qa)}{iQ + q\tan(Qa)}. \end{aligned}$$

Recalling $S_1 = e^{-2iqa}$, we may factorize $S = S_2S_1$ with

$$\boxed{S_2 = \frac{iQ - q\tan(Qa)}{iQ + q\tan(Qa)}}.$$

The two expressions for S_2 are equivalent wherever their individual trigonometric denominators are defined; the matching result has a regular continuation at the remaining values.

Note. Here $|S| = 1$: the potential is real, and the infinite wall permits no transmitted flux. The quantity S changes the *phase* of the reflected wave, not its probability of reflection.

(8)

Introduce the parameters given in the question,

$$U_0 \equiv \frac{2mV_0}{\hbar^2}, \qquad \epsilon \equiv \frac{q}{\sqrt{U_0}} \ll 1, \qquad \theta \equiv \sqrt{U_0}\, a.$$

Since

$$Q = \sqrt{q^2 + U_0} = \sqrt{U_0}\sqrt{1+\epsilon^2} = \sqrt{U_0}\big[1 + O(\epsilon^2)\big],$$

we also have $\tan(Qa) = \tan\theta + O(\epsilon^2)$, provided $\sin\theta \neq 0$ and $\cos\theta \neq 0$. Thus,

$$\begin{aligned} S_2 &= \frac{iQ - q\tan(Qa)}{iQ + q\tan(Qa)} \\ &\simeq \frac{i - \epsilon\tan\theta}{i + \epsilon\tan\theta} \\ &= \frac{1 + i\epsilon\tan\theta}{1 - i\epsilon\tan\theta} \\ &\simeq 1 + 2i\epsilon\tan\theta. \end{aligned}$$

Therefore, to first order in ϵ,

$$\boxed{S_2 \simeq 1 + 2i\epsilon\tan\big(\sqrt{U_0}a\big)}.$$

(9)

Now impose the special condition

$$\tan\big(\sqrt{U_0}a\big) = \sqrt{U_0}a.$$

The first-order correction in (8) is essential:

$$\begin{aligned} S_2 &\simeq 1 + 2i\epsilon\sqrt{U_0}a \\ &= 1 + 2iqa, \end{aligned}$$

where $\epsilon\sqrt{U_0} = q$ has been used. Meanwhile, the condition $qa \ll 1$ gives

$$S_1 = e^{-2iqa} \simeq 1 - 2iqa.$$

Multiplying and discarding terms of second order or higher,

$$\begin{aligned} S = S_2 S_1 &\simeq (1 + 2iqa)(1 - 2iqa) \\ &\simeq \boxed{1}. \end{aligned}$$

Hence, in the region $x > a$,

$$\phi_2(x) = e^{-iqx} - Se^{iqx} \simeq e^{-iqx} - e^{iqx} = \phi_0(x).$$

Consequently,

$$\boxed{f_2 = |\phi_0(x) - \phi_2(x)|^2 \simeq 0}.$$

Physical interpretation. At this special well depth and width, the low-energy phase shift produced by the attractive well cancels the phase shift due to the region of width a. The reflected wave therefore approaches that of the original wall at $x = 0$: the added potential has almost no effect on the exterior wavefunction at the stated order.